UNVEILING ACADEMIC TRAJECTORIES THROUGH MACHINE LEARNING ANALYSIS

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Abstract

Our goal is to predict student academic trajectories—specifically dropout, retention, and graduation outcomes—using machine learning models trained on academic and demographic data.

We applied several classification algorithms including Random Forest, SVM, LightGB, HistGBM and Logistic Regression.

We evaluate model performance using accuracy, precision, recall, F1-score, and AUC.

Our results show that tree-based ensemble models outperform linear models, with LightGBM and Random Forest achieving F1-scores above 0.90 and AUC up to 0.913.

Datasets

- Student academic dataset from Kaggle
- 37 features across academic, demographic, and socioeconomic dimensions
- Target label (graduate, dropout, enrolled)
- 4424 total samples

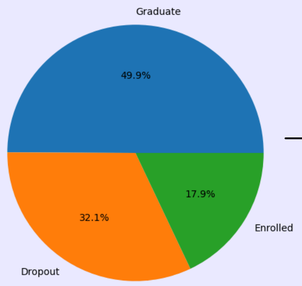
Methodology

The methodology consists of two main stages: **Data Preparation and Modelling** & **Validation Process**. In the data preparation stage, relevant academic and demographic features were **collected** and **explored** to understand their distributions and relationships. **Feature analysis** was conducted using correlation matrices and domain understanding to identify the most relevant predictors. The data was then **preprocessed** through encoding of categorical variables, scaling of numerical features, and handling of class imbalance using SMOTE.

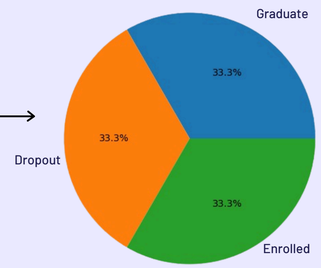
In the modelling and validation phase, the **dataset was split into training and testing subsets.** Models were trained using various algorithms, and class balancing was applied only on the training data to prevent information leakage. After training, **model performance was validated using cross-validation**, After that, we used RandomizedSearchCV to find the best hyperparameters for each model and the best-performing model was selected based on accuracy, F1-score, and AUC. The **final evaluation** confirmed the model's ability to predict student dropout, retention, and graduation accurately.

Results

Distribution of Student Academic Status



The target distribution shows an imbalance in the data, with the majority of students in the Graduate category. This is a concern because it can affect the performance of the classification model, especially in detecting the more critical Dropout class.



SMOTE successfully balances the target distribution, giving the model a fairer chance to learn to detect all categories of academic status.

Models

Logistic Regression

$$P(y=1|x) = \frac{1}{1+e^{-(\beta_0+\beta_1 x_1+\beta_2 x_2+\cdots+\beta_n x_n)}}$$

Support Vector Machine

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}\xi_i \quad \text{s.t. } y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1-\xi_i$$

Histogram-based Gradient Boosting Machine

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x)$$

Random Forest

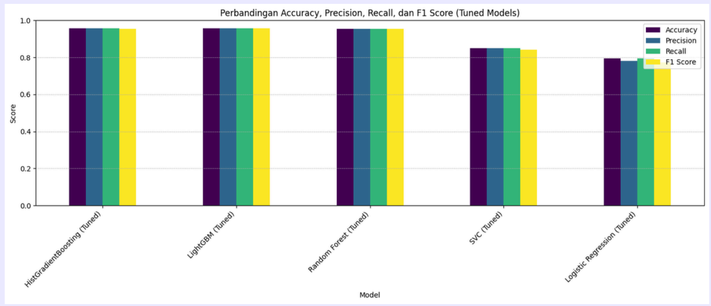
$$f(x) = \frac{1}{B}\sum_{b=1}^{B} f_b(x)$$

Light Gradient Boosting Mechine

$$V_{j|O}(d) = \frac{1}{n_o}\left(\frac{\left(\sum_{\{x_i \in O: x_{ij} \leq d\}} g_i\right)^2}{n^l_{j|O}(d)} + \frac{\left(\sum_{\{x_i \in O: x_{ij} > d\}} g_i\right)^2}{n^l_{r|O}(d)}\right)$$

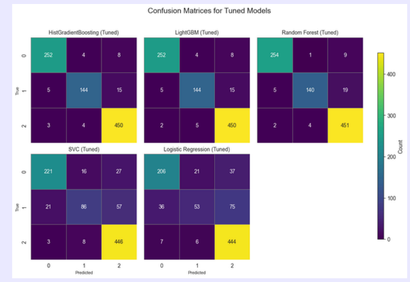
Comparison of Model Evaluation

Rank	Model	Accuracy	Precision	Recall	F1 Score
1	HistGradientBoosting	0.955932	0.955959	0.955932	0.955537
2	LightGBM (Tuned)	0.955932	0.955927	0.955932	0.955553
3	Random Forest	0.954802	0.955424	0.954802	0.954191
4	SVC (Tuned)	0.850847	0.848505	0.850847	0.842047
5	Logistic Regression	0.79435	0.781923	0.79435	0.772743



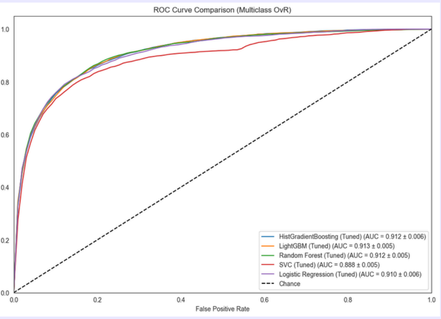
Based on the evaluation results, the HistGradientBoosting (Tuned) and LightGBM (Tuned) models showed the best performance with very high and almost identical Accuracy, Precision, Recall, and F1 Score values, which were around 0.9559. This shows that both models are very consistent and reliable in classifying data correctly. Random Forest also provides competitive performance, slightly below the top two models. Meanwhile, SVC (Tuned) and Logistic Regression (Tuned) have much lower performance, especially Logistic Regression which has the lowest F1 Score (0.7727), indicating that this model is less suitable for this dataset than other models. Overall, ensemble models such as HistGradientBoosting and LightGBM prove to be superior in accuracy and balance between precision and recall.

Confusion Matrix



Based on the confusion matrices, HistGradientBoosting, LightGBM, and Random Forest demonstrate strong classification performance with high accuracy and minimal misclassifications across all classes—particularly excelling in classifying class 2 correctly. In contrast, SVC and Logistic Regression struggle notably with class 1, often misclassifying it as other classes. Overall, HistGradientBoosting and LightGBM emerge as the most reliable models, while Logistic Regression shows the weakest performance and is the least suitable for this multiclass classification task.

ROC Curve



The ROC curve comparison shows that LightGBM, HistGradientBoosting, and Random Forest achieve the highest AUC scores (around 0.912–0.913), indicating excellent discriminative ability across all classes. Logistic Regression follows closely with an AUC of 0.910, performing slightly below the top three but still demonstrating good classification ability. SVC has the lowest AUC (0.888), indicating weaker performance in separating the classes compared to the other models. Overall, LightGBM slightly leads, but the top three models are nearly tied in terms of ROC-based evaluation.

Conclusion

Tree-based models such as HistGradientBoosting, LightGBM, and Random Forest demonstrated strong performance in predicting student dropout, retention, and graduation outcomes, even in the presence of class imbalance. These models outperformed linear classifiers in terms of accuracy, precision, recall, and F1-score.
Key predictive features included students' GPA, credit-related variables, age at enrollment, admission grades, and scholarship status—emphasizing the critical role of early academic performance and socioeconomic background in shaping academic trajectories.

References

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